DATE:10/10/2026

CO1-ASSESSMENT TOOL – 1

APPLICATION BASED QUESTIONS

COURSE CODE : DSA0308

COURSE NAME: Natural Language Processing

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1: Resume Information Extraction

A. Simulation and Demonstration

A multinational recruitment company receives thousands of resumes. The system extracts candidate details and checks eligibility automatically using Regular Expressions.

Sample input

- Candidate Name: Manoj Kumar
- Email: manojkumar@gmail.com
- Mobile Number: 9391843022
- Technical Skills: Python, Java, SQL, Machine Learning, NLP
- Experience: 3 years

Demonstration output

Resume Information Extraction Result

Candidate Name: Manoj Kumar

Email Address: manojkumar@gmail.com

Mobile Number: 9391843022

Technical Skills: Python, Java, SQL, Machine Learning, NLP

Years of Experience: 3

Profile Summary: Candidate has 3 years of experience in the listed technical skills.

Eligibility: Eligible

B. Step-by-Step Working and Analysis

Step 1: Input collection

The resume is provided as text containing the candidate's personal and professional information.

Step 2: Information extraction

Regular Expressions identify the email address and mobile number by matching their patterns.

Step 3: Skill identification

The system checks the resume for technical skills such as Python, Java, SQL, Machine Learning, and NLP.

Step 4: Experience analysis

The system identifies three years of experience from the resume.

Step 5: Eligibility checking

The system checks whether the candidate has at least two years of experience and at least one listed technical skill.

Step 6: Result interpretation

The candidate meets both example conditions. Therefore, the system displays the candidate's extracted details and marks the candidate as eligible.

Result: The resume information extraction system successfully extracts the required information, generates a candidate summary, and checks eligibility.

2: Product Search System

A. Simulation and Demonstration

An online shopping company wants to help customers find products easily by searching with keywords.

Sample Product List:

1. Python Programming Book
2. Java Programming Book
3. SQL Database Guide
4. Machine Learning Course
5. NLP Basics Book
6. Python Notebook

Search Input:

Keyword: Python

Search Type: Prefix Search

Output:

Product Search Result

Search Keyword: Python

Search Type: Prefix

Matching Products:

1. Python Programming Book
2. Python Notebook

Total Matching Products: 2

Other Search Results:

Exact Search: Python Notebook — 1 match

Prefix Search: Python — 2 matches

Suffix Search: Book — 3 matches

Partial Search: learning — 1 match

Case-insensitive Search: PYTHON — 2 matches

B. Step-by-Step Working and Analysis

Step 1: Product List

The system stores all product names in a list.

Step 2: Enter Keyword

The user enters a word to search for, such as Python.

Step 3: Exact Search

The system finds a product with the same complete name.

Step 4: Prefix Search

The system finds products whose names start with the given word.

Step 5: Suffix Search

The system finds products whose names end with the given word.

Step 6: Partial Search

The system finds products that contain the given word anywhere in the name.

Step 7: Case-insensitive Search

The system treats capital and small letters as the same. For example, Python and PYTHON give the same results.

Step 8: Result Analysis

The system displays the matching products and counts them. When the keyword is Python and the search type is Prefix, two products are found.

Result: The product search system finds products using different search methods and displays the matching products and their total count.

Question 3: University Registration System

A. Simulation and Demonstration

A university wants to check student details before allowing students to register for a course.

Sample Input:

Register Number: 24AI123

Email: manoj@simats.edu.in

Course Code: CSE101

Semester: 3

Mobile Number: 9876543210

Output:

University Registration Result

Register Number: Valid

Email: Valid

Course Code: Valid

Semester: Valid

Mobile Number: Valid

Final Registration Status: SUCCESSFUL

B.Step-by-Step Working and Analysis

Step 1: Enter Student Details

The system receives the student's register number, email, course code, semester, and mobile number.

Step 2: Check Register Number

The system checks whether the register number follows the required format.

Step 3: Check Email

The system checks whether the email address follows the university's email format.

Step 4: Check Course Code

The system checks whether the course code is correct.

Step 5: Check Semester

The system checks whether the semester number is within the allowed range.

Step 6: Check Mobile Number

The system checks whether the mobile number has the correct format.

Step 7: Check All Details

The system checks whether all the details are valid. If all are valid, registration is successful. Otherwise, registration fails.

Step 8: Result Analysis

In this example, all five details are valid according to the sample rules. Therefore, the system displays successful.

Result: The university registration system checks student details and shows whether the registration is successful or not.